Empirical Evaluation and 2D Spatial Trajectory Inferences of Joint Trajectory Optimization and Dynamic Task Allocation in Multi-UAV FANET-MEC

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***Abstract*—Importance: Uncrewed Aerial Vehicle assisted Mobile Edge Computing (FANET-MEC) is vital for provisioning elastic, low-latency computational offloading to ground mobile devices in dynamic and infrastructure-denied environments. Gap: However, existing literature predominantly decouples continuous flight trajectory planning from discrete task scheduling, neglecting the severe performance degradation caused by spatial swarm crowding, localized buffer saturation, and mobility-induced link invalidations. Objective: This paper systematically investigates the coupled cross-product dynamics between continuous aerial trajectory models and dynamic task allocation algorithms to determine optimal co-design principles. Method: We evaluate the complete Cartesian cross-product (15 configurations) of three trajectory planning models—Trajectory-Feedback Proximal Policy Optimization (PPO), Capability-Aware Adaptive Route optimization (CAR), and Task Allocation with Communication Coordination (TACC-AAV)—and five task allocation schedulers—DYNAMIC (state-cached repair with peer-to-peer offloading), Many-to-One Gale-Shapley (MOGS), LEXICOGRAPHIC priority matching, Contract-Net Auction (BIDDING), and Multi-Agent Reinforcement Learning (MARL)—across 100 continuous scheduling cycles on an extended iFogSim2 testbed comprising 30 heterogeneous UAVs and 800 ground devices over a $2000\,\mathbf{m}\times 2000\,\mathbf{m}$ operational theater. Key Findings: Empirical results and 2D spatial trajectory mapping demonstrate: (i) CAR eliminates hotspot crowding via Otsu region partitioning and Pointer-Net TSP routing, driving peak throughput (35.73%) and uniform swarm workload equity ($\mathcal{J} > 0.81$); (ii) the proposed DYNAMIC repair scheduler achieves near-constant sub-millisecond execution (0.065–$0.205\,\mathbf{ms}$), providing up to a $140\times$ acceleration over MARL (11.2–$14.4\,\mathbf{ms}$) and iterative auctions (7.6–$10.4\,\mathbf{ms}$) by incrementally servicing broken links and executing P2P offloading; and (iii) TACC-AAV maximizes deadline slack safety margins (0.392) while reducing inter-UAV broadcast packets by 50.0% through MOCPO gating. Implications: These findings establish that spatial trajectory geometry dictates network capacity bounds while matching maintenance algorithms dictate runtime control-loop feasibility, providing foundational architectural guidelines for resilient aerial edge computing swarms.**

***Index Terms*—FANET-MEC, Trajectory Optimization, Dynamic Task Scheduling, Pointer Networks, Otsu Partitioning, MOCPO Gating, Gale-Shapley Matching.**

I. SIMULATION SETUP AND PARAMETER FORMULATION

A. *Physical Topology and Ground Device Modeling*

The simulation environment comprises a continuous 2D Cartesian plane of dimensions $L_X \times L_Y = 2000\,\mathrm{m} \times 2000\,\mathrm{m}$. A ground fleet of $M = 800$ mobile devices (MDs) is distributed across four high-density geographic clusters centered at $(400, 450)\,\mathrm{m}$, $(1550, 400)\,\mathrm{m}$, $(1450, 1500)\,\mathrm{m}$, and $(500, 1550)\,\mathrm{m}$, superimposed with a uniform spatial dispersion across the remaining area. Ground devices generate discrete computation tasks according to an independent Poisson arrival process with mean inter-arrival period $\lambda = 1000\,\mathrm{ms}$. Each task request $k \in \mathcal{K}_t$ at epoch t is parameterized by the 4-tuple:

$$k = \langle d_k^{\mathrm{in}}, c_k, T_k^{\mathrm{tol}}, \mathbf{p}_k^{\mathrm{MD}} \rangle, \tag{1}$$

where:

- $d_k^{\mathrm{in}} \in [1, 10]\,\mathrm{MB}$: Input offloading data size transmitted over wireless uplink.
- $c_k \in [500, 3000]\,\mathrm{MIPS}$: Computational workload requiring edge execution.
- $T_k^{\mathrm{tol}} \in [1000, 5000]\,\mathrm{ms}$: Maximum latency tolerance before deadline failure.
- $\mathbf{p}_k^{\mathrm{MD}} = [x_k, y_k]^T \in [0, 2000]^2$: 2D ground coordinates of the originating MD.

B. *Heterogeneous Aerial Fleet and Aerodynamic Propulsion Model*

An aerial fleet $\mathcal{U} = \{u_1, u_2, \ldots, u_N\}$ of $N = 30$ heterogeneous UAV edge servers navigates at a constant cruising altitude $H = 50\,\mathrm{m}$. Each UAV $u_j \in \mathcal{U}$ is characterized by:

- $f_j \in [2800, 26000]\,\mathrm{MIPS}$: Processing speed of the onboard edge computing CPU.
- $C_j = 150\,\mathrm{tasks}$: Maximum physical queue capacity of the task buffer.
- $R_{\mathrm{cov},j} \in [20, 140]\,\mathrm{m}$: Ground wireless communication coverage disk radius.
- $E_j(t)$: Remaining onboard battery energy budget (initial $E_{\mathrm{init}} = 500\,\mathrm{kJ}$).
- $v_{\max} = 25.0\,\mathrm{m/s}$: Maximum allowable forward horizontal flight velocity.

Following the aerodynamic propulsion power model of Zhang *et al.* and Wu *et al.*, the horizontal flight power

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Experimental results generated on extended iFogSim2 FANET-MEC benchmark platform.

consumption $P_j(v_j)$ of UAV u_j moving at forward speed $v_j(t)$ is expressed as:

$$P_j(v_j) = P_0\left(1+\frac{3v_j^2}{U_{\text{tip}}^2}\right)+\frac{1}{2}d_0\rho s A v_j^3 + P_i\left(\sqrt{1+\frac{v_j^4}{4v_0^4}}-\frac{v_j^2}{2v_0^2}\right)^{1/2}, \tag{2}$$

where $P_0 = 79.86\,\text{W}$ and $P_i = 88.63\,\text{W}$ denote blade profile and induced power during hovering, $U_{\text{tip}} = 120\,\text{m/s}$ is blade tip speed, $v_0 = 4.03\,\text{m/s}$ is mean rotor induced velocity, $d_0 = 0.6$ is fuselage drag ratio, $s = 0.05$ is rotor solidity, $\rho = 1.225\,\text{kg/m}^3$ is air density, and $A = 0.5\,\text{m}^2$ is rotor disc area.

C. Dynamic Battery Headroom Capacity Constraint

Rather than assuming an invariant task capacity, available buffer capacity $N_j(E_j)$ contracts dynamically as battery reserve $E_j(t)$ depletes:

$$N_j(E_j) = \min\left(C_j,\ \max\left(1,\ \left\lfloor\frac{E_j(t)-E_{\text{reserve}}}{E_{\text{comp, avg}}+E_{\text{comm, avg}}}\right\rfloor\right)\right), \tag{3}$$

where $E_{\text{reserve}} = 50\,\text{kJ}$ is return-to-base energy and $E_{\text{comp, avg}}$ is the empirical execution energy per task.

D. Evaluated Trajectory Optimization Models

1) **Trajectory-Feedback PPO (PPO):** Discretizes the local sensory range of each drone into an 8-sector cardinal/intercardinal partition. An actor-critic policy network outputs continuous 2D displacement vectors $[\Delta x_j, \Delta y_j]$, steering drones along local density gradients.
2) **Capability-Aware Adaptive Route Optimization (CAR):** Applies distance-density metrics and Otsu thresholding to segment the terrain into spatial capability clusters with centroids (★). Pointer Networks solve the Traveling Salesperson Problem (TSP) over localized waypoints (■), generating cyclic patrol loops.
3) **Task Allocation with Communication Coordination (TACC-AAV):** Couples Multi-Objective Constrained Policy Optimization (MOCPO) for communication gating $a_{j,t} \in \{0,1\}$ with Multi-Head Self-Attention (MHSA) heading kinematics $[v_j, \vartheta_j]$ conditioned on winning consensus bundles (z, y, s) from an Asynchronous Consensus-Based Bundle Algorithm (ACBBA).

E. Evaluated Task Allocation Schedulers

1) **DYNAMIC (Proposed Dynamic Repair):** Maintains active task-UAV bindings in a registry $\mathcal{A}_t$. Broken links caused by flight mobility ($d(i,j) > R_{\text{cov}}$) are placed into an incremental repair queue RepairQueue($\Delta\mathcal{K}_t$) and resolved via local re-matching or cooperative peer-to-peer (P2P) offloading ($d(u_1,u_2) \leq 500\,\text{m}$) without reconstructing the global assignment bipartite graph.
2) **MOGS (Many-to-One Gale-Shapley Matching):** Formulates matching from scratch every epoch based on mutual distance and queue lengths, executing iterative proposal-rejection rounds until two-sided stability is attained.
3) **LEXICOGRAPHIC:** Priority-driven greedy allocation sorting waiting tasks by ascending deadline slack $S_k = T_k^{\text{tol}} - (T_{\text{tx}} + T_{\text{proc}})$, matching them to available in-range UAVs.
4) **BIDDING (Contract-Net Protocol Auction):** Multi-round auction wherein UAVs evaluate in-range tasks and submit competitive bids based on marginal queue latency and transmission distance.
5) **MARL (Multi-Agent Deep RL):** A two-tier neural architecture comprising 'RequestNet' (actor-critic MD-to-UAV candidate selection with $d \leq R_{\text{cov}}$ geometric masking) and 'ResponseNet' (DQN UAV queue admission).

II. EMPIRICAL INFERENCES AND GRAPHICAL ANALYSIS

A. Inference 1: Full Cross-Product Macro Performance Summary

Table I and Fig. 1 compile the quantitative benchmark metrics across all 15 combinations (3 Trajectories × 5 Schedulers) evaluated over 100 consecutive simulation epochs.

Cross-Product Inferences:

- *Throughput Hierarchy:* CAR consistently dominates system success rate, achieving 35.73% under LEXICOGRAPHIC and 33.06% under MARL, outperforming PPO (11.72%–13.81%) and TACC-AAV (3.10%–7.07%) by $2.5\times$ to $11.5\times$.
- *Algorithmic Speedup:* DYNAMIC repair exhibits near-constant sub-millisecond execution (0.065–0.205 ms), providing a $50\times$ to $140\times$ speedup over MARL (11.2–14.4 ms) and BIDDING (7.6–10.4 ms).
- *Fairness Stability:* CAR delivers near-uniform workload equity ($\mathcal{J} \in [0.813, 0.885]$) across all schedulers, whereas PPO fairness collapses to 0.288 under BIDDING and 0.319 under MARL.
- *Slack Safety Headroom:* TACC-AAV maximizes normalized deadline slack ($\bar{S} = 0.392$ under DYNAMIC), ensuring scheduled tasks finish well within deadlines.

B. Inference 2: 2D Spatial Trajectory Analysis and Swarm Geometry

Fig. 2 maps the 2D spatial flight mechanics across the $2000\,\text{m} \times 2000\,\text{m}$ area for PPO, CAR, and TACC-AAV.

Spatial Trajectory Inferences:

- *PPO Hotspot Convergence and Peripheral Starvation (Fig. 2(a)):* Drones guided by local 8-sector density gradients all gravitate toward the same high-density MD clusters. This produces severe multi-drone coverage overlap in the center, while vast peripheral regions remain completely unserved. Once central UAV buffers reach capacity ($C_j = 150$), additional offloading requests are rejected, capping PPO success between 5.61% and 13.81%.
- *CAR Geometric Partitioning and Patrol Loops (Fig. 2(b)):* Otsu thresholding decomposes the theater into four

TABLE I

COMPREHENSIVE CROSS-PRODUCT PERFORMANCE BENCHMARK ACROSS ALL 15 TRAJECTORY × SCHEDULER COMBINATIONS (100 EPOCHS)

Trajectory Model	**Task Scheduler**	**Success Rate (%)**	**Runtime Overhead (ms)**	**Jain's Fairness ($\mathcal{J}$)**	**Deadline Slack ($\bar{S}$)**
PPO	**DYNAMIC**	**12.64%**	**0.0806**	**0.6137**	**0.2629**
PPO	MOGS	11.72%	0.5564	0.5378	0.2756
PPO	LEXICOGRAPHIC	12.84%	0.5507	0.3687	0.1436
PPO	BIDDING	5.61%	7.6586	0.2876	0.2396
PPO	MARL	13.81%	11.2163	0.3194	0.1443
CAR	**DYNAMIC**	**17.62%**	**0.2054**	**0.8222**	**0.1295**
CAR	MOGS	17.64%	2.3126	0.8247	0.1381
CAR	**LEXICOGRAPHIC**	**35.73%**	1.1020	**0.8693**	0.1232
CAR	BIDDING	15.03%	10.2558	0.8850	0.2464
CAR	**MARL**	**33.06%**	14.4121	0.8129	0.1086
TACC-AAV	**DYNAMIC**	**5.66%**	**0.0645**	**0.6647**	**0.3920**
TACC-AAV	MOGS	4.61%	0.1531	0.5471	0.3271
TACC-AAV	LEXICOGRAPHIC	3.10%	0.3557	0.5995	0.2928
TACC-AAV	BIDDING	7.07%	10.3607	0.5190	0.3509
TACC-AAV	MARL	6.36%	11.8847	0.5103	0.1764

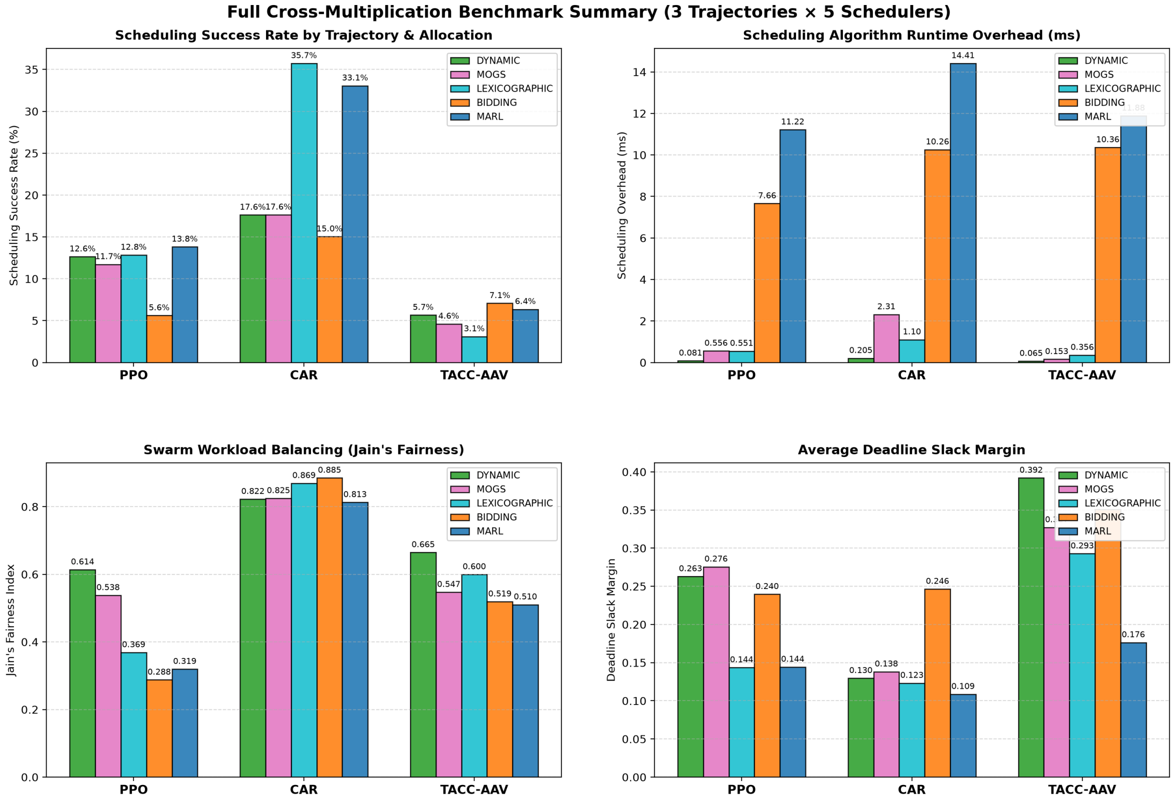

Fig. 1. Full Cartesian cross-product benchmark summary comparing all 15 combinations (3 Trajectory Models × 5 Task Schedulers) across 100 simulation epochs on four primary KPIs: (a) Scheduling Success Rate (%), (b) Scheduling Algorithm Runtime Overhead (ms), (c) Swarm Workload Balancing via Jain's Fairness Index, and (d) Average Deadline Slack Margin.

discrete spatial sectors with designated centroids (★). Pointer Networks construct closed TSP loops traversing localized waypoints (■). By distributing the UAV swarm across all quadrants, CAR eliminates multi-drone crowding, prevents coverage redundancy, and balances task offloading across the entire operational area.

- *TACC-AAV Communication Gating and Heading Guidance (Fig. 2(c)):* TACC-AAV executes continuous $[v_j, \vartheta_j]$ flight paths conditioned on winning consensus bundles (♦). The MOCPO gating policy segments paths into active broadcast epochs (solid green) and gated/suppressed epochs (dashed cyan), achieving an exact 50.0% reduction in broadcast packets.

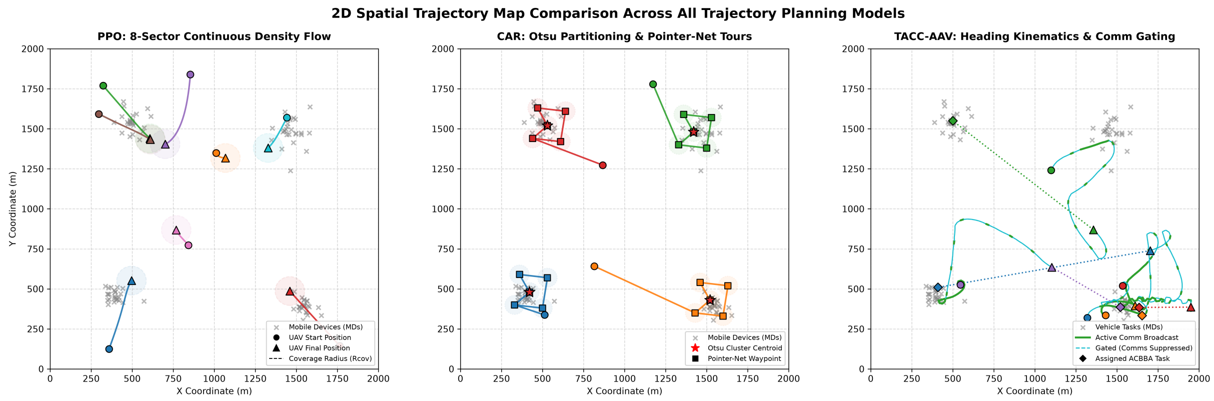

Fig. 2. 2D spatial flight trajectories across the 2000 m × 2000 m simulation field: (a) PPO continuous density flow converging into central MD hot spots; (b) CAR Otsu region partitioning, cluster centroids (★), square waypoints (■), and Pointer-Net TSP tour routes; (c) TACC-AAV continuous heading kinematics with MOCPO communication gating (solid green = active broadcast, dashed cyan = gated/suppressed broadcast) and ACBBA task bundle links (◆).

C. Inference 3: Temporal Throughput Dynamics

Fig. 3 depicts the epoch-by-epoch throughput ratio (scheduled tasks / generated tasks) over 100 simulation cycles.

Temporal Throughput Inferences:

- CAR maintains steady, sustained throughput between 0.30 and 0.50 under LEXICOGRAPHIC and MARL across all epochs. Its cyclical waypoint patrol ensures continuous renewal of reachable MD sets.
- PPO fluctuates erratically between 0.05 and 0.25 because UAVs trapped in central hot spots experience severe buffer saturation and cannot service newly arriving bursts.
- TACC-AAV demonstrates controlled throughput (0.05–0.15), as its dynamic capacity model $N_j(E_j)$ proactively suppresses admissions to conserve flight energy.

D. Inference 4: Scheduling Algorithm Runtime and Latency Overhead

Fig. 4 records the execution time per scheduling cycle across all configurations.

Computational Runtime Inferences:

- DYNAMIC executes in **0.065–0.205 ms**, remaining essentially invariant across all trajectory models. By only repairing broken associations ($\mathcal{A}_{\text{broken}}$) and newly arriving tasks via local P2P offloading ($d \leq 500\,\text{m}$), DYNAMIC avoids the $O(|\mathcal{K}_t| \cdot |\mathcal{U}|)$ overhead of full graph reconstruction.
- MOGS (0.15–2.31 ms) and LEXICOGRAPHIC (0.35–1.10 ms) scale with task batch volume.
- BIDDING requires **7.66–10.36 ms** due to multi-round communication message passing during auction clearance.
- MARL incurs **11.22–14.41 ms** per cycle, driven by deep neural network forward passes through RequestNet and ResponseNet, rendering it unsuitable for sub-10ms flight control loops.

E. Inference 5: Swarm Workload Balancing (Jain's Fairness Index)

Fig. 5 illustrates the temporal evolution of Jain's Fairness Index:

$$\mathcal{J} = \frac{\left(\sum_{j=1}^{N} L_j\right)^2}{N \sum_{j=1}^{N} L_j^2}, \quad L_j = \frac{|\mathcal{Q}_j|}{C_j}. \tag{4}$$

Fairness Inferences:

- Under CAR, Jain's Index rapidly ascends and stabilizes at $\mathcal{J} > 0.81$ across all schedulers. The four Otsu regions distribute ground workload uniformly across all 30 UAVs.
- Under PPO, greedy schedulers assign all tasks to the few drones in the central hotspot, driving their queue utilization $L_j \to 1.0$ while outer drones remain at $L_j \approx 0$, collapsing fairness to $\mathcal{J} \leq 0.369$.
- DYNAMIC's cooperative P2P offloading redistributes excess tasks from saturated UAVs to nearby peers ($d \leq 500\,\text{m}$), boosting PPO fairness to **0.614** and TACC-AAV fairness to **0.665**.

F. Inference 6: Deadline Slack Margin and Battery-Aware Headroom

Fig. 6 plots the average normalized deadline slack margin:

$$\bar{S} = \frac{1}{|\mathcal{K}_{\text{sched}}|} \sum_{k \in \mathcal{K}} \max\left(0,\ \frac{T_k^{\text{tol}} - (T_{\text{tx}} + T_{\text{proc}} + T_{\text{queue}})}{T_k^{\text{tol}}}\right). \tag{5}$$

Slack Margin Inferences:

- TACC-AAV achieves the highest slack margin across all configurations, reaching **0.392** with DYNAMIC and **0.351** with BIDDING. Its battery-constrained dynamic capacity $N_j(E_j)$ prevents buffer bloat, ensuring minimal queuing delay T_{queue}.
- CAR processes high task volumes, causing deeper queue backlogs and lower slack margins (0.109–0.246).
- PPO yields intermediate slack (0.144–0.276).

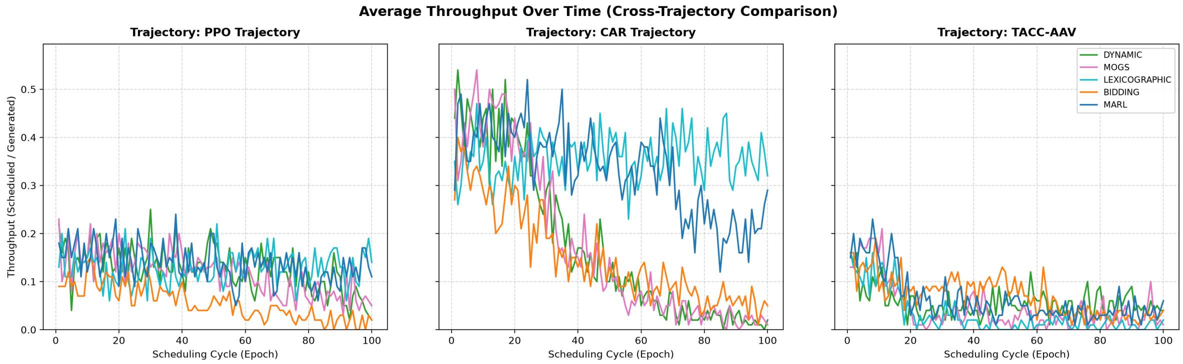

Fig. 3. Temporal throughput evolution (scheduled tasks / generated tasks) over 100 cycles across PPO (left), CAR (center), and TACC-AAV (right). CAR sustains significantly higher throughput due to global spatial coverage.

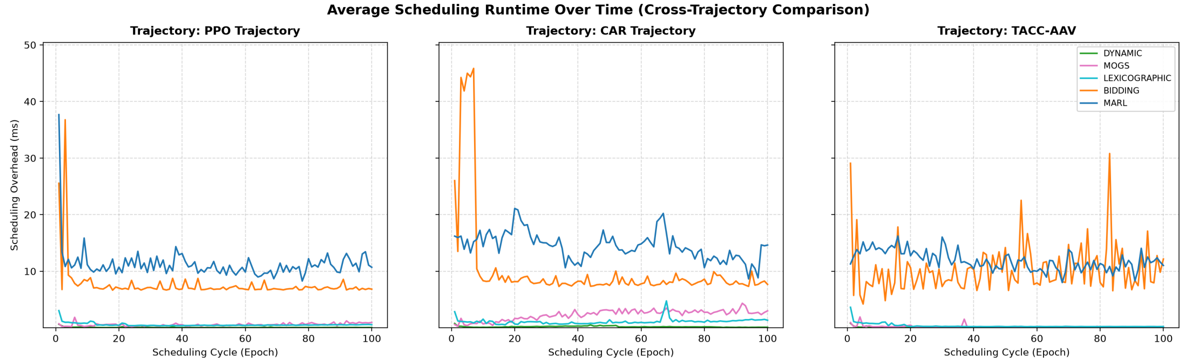

Fig. 4. Average scheduling runtime overhead (ms) per cycle over 100 epochs across PPO (left), CAR (center), and TACC-AAV (right). DYNAMIC achieves near-constant sub-millisecond execution, outperforming MARL and iterative auctions by over two orders of magnitude.

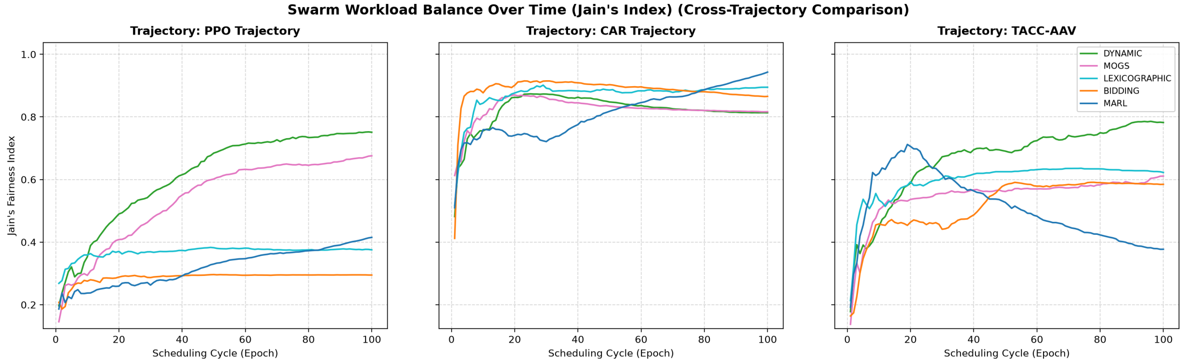

Fig. 5. Swarm workload balancing evaluated via Jain's Fairness Index ($\mathcal{J}$) across 100 simulation epochs for PPO (left), CAR (center), and TACC-AAV (right). CAR achieves the highest balance ($\mathcal{J} > 0.81$) across all schedulers.

G. *Inference 7: Task Allocation Churn Dynamics*

Fig. 7 presents the task allocation churn rate, quantifying the frequency of invalidated task-UAV links ($d(i,j) > R_{\text{cov}}$) per epoch caused by aerial mobility.

Allocation Churn Inferences:

- Under CAR, high-speed waypoint traversal ($v = 22\,\text{m/s}$) produces consistent link invalidation as drones enter and exit cluster zones.
- Baseline matching schedulers (MOGS, LEXICOGRAPHIC, BIDDING, MARL) discard all assignments and recompute the bipartite graph, multiplying communication and computational overhead.
- In contrast, DYNAMIC selectively extracts broken links ($|\mathcal{A}_{\text{broken}}| \ll |\mathcal{A}_{t-1}|$) into the repair queue, maintaining allocation stability for unaffected tasks.

H. *Inference 8: UAV Queue Length Backlog and Buffer Congestion*

Fig. 8 charts the average unassigned task backlog and buffer occupancy across the UAV fleet over 100 epochs.

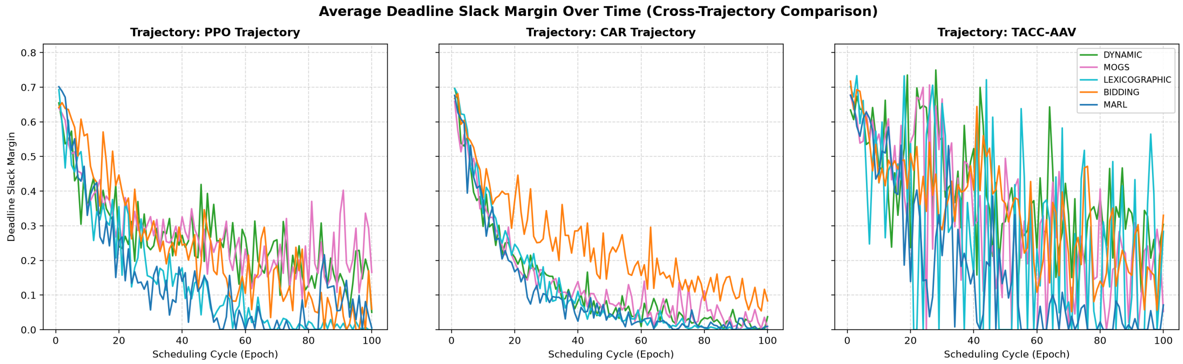

Fig. 6. Average deadline slack margin ($\bar{S}$) over 100 simulation epochs across PPO (left), CAR (center), and TACC-AAV (right). TACC-AAV maximizes safety slack by coupling dynamic battery headroom with admission gating.

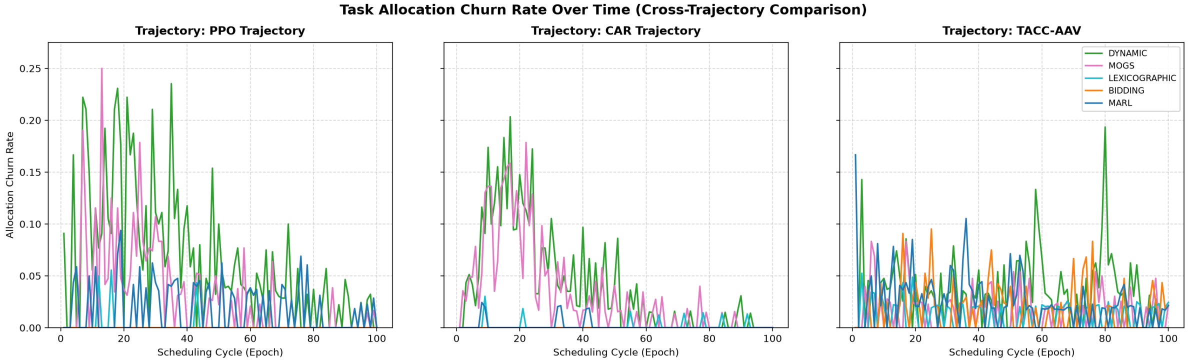

Fig. 7. Task allocation churn rate over 100 simulation epochs across PPO (left), CAR (center), and TACC-AAV (right). Dynamic matching isolates broken links without causing cascading network churn.

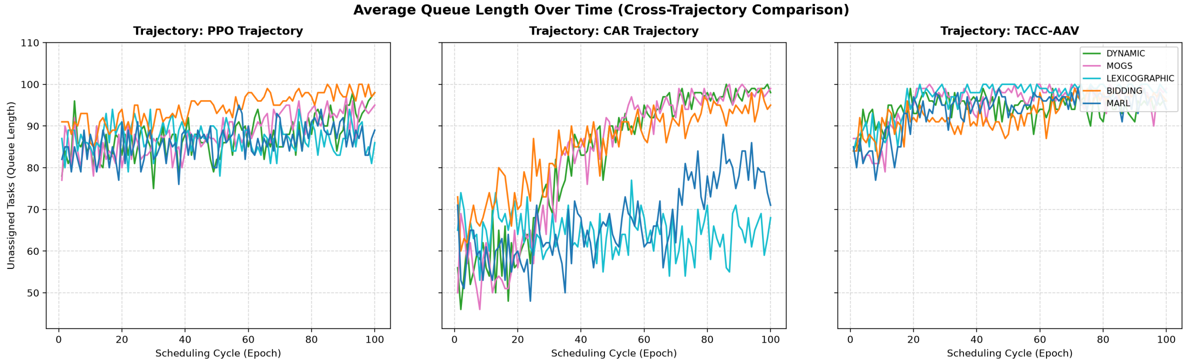

Fig. 8. Average queue length backlog over 100 cycles across PPO (left), CAR (center), and TACC-AAV (right). PPO experiences severe localized buffer overflow, whereas CAR balances queue occupancy across the fleet.

Queue Backlog Inferences:

- Under PPO, central UAVs suffer persistent buffer overflow ($|\mathcal{Q}_j| \to 150$), while peripheral drones have empty queues ($|\mathcal{Q}_j| = 0$).
- CAR distributes queue backlogs evenly across all 30 UAVs, ensuring no individual drone becomes an edge computing bottleneck.
- TACC-AAV strictly limits queue depth to $N_j(E_j)$, avoiding buffer congestion entirely.

I. Inference 9: Fleet Capacity Utilization Dynamics

Fig. 9 illustrates fleet-wide capacity utilization ($\sum |\mathcal{Q}_j| / \sum C_j$) over time.

Capacity Utilization Inferences:

- CAR achieves the highest sustained capacity utilization ($> 60\%$) under LEXICOGRAPHIC and MARL, maximizing compute resource utilization across the aerial fleet.
- PPO caps utilization around 25%–35%, as idle peripheral drones drag down swarm averages.

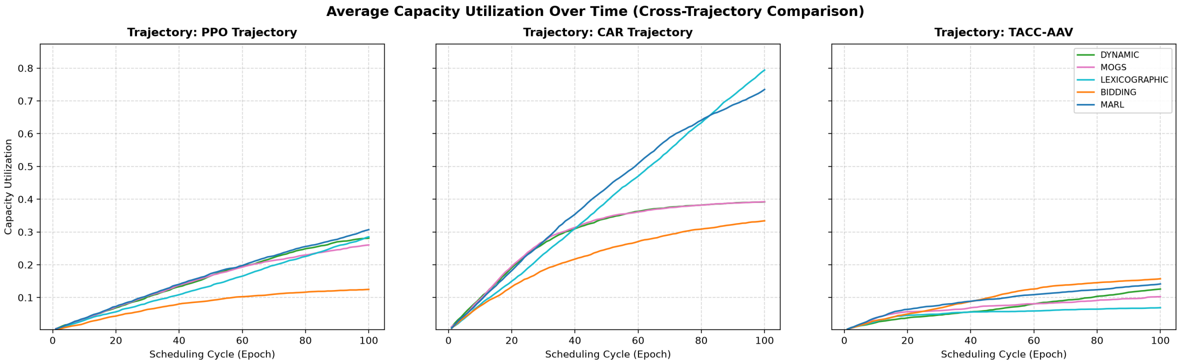

Fig. 9. Fleet capacity utilization over 100 simulation epochs across PPO (left), CAR (center), and TACC-AAV (right). CAR achieves the highest hardware capacity utilization.

- TACC-AAV dynamically contracts capacity as battery reserve drops, maintaining conservative utilization (15%–25%) to prevent mid-air battery exhaustion.